

SHAP the easy way (linear model) — worksheet

OPIM 5512 · Data Science Using Python · Dr. Dave Wanik — name: _____

SHAP's "fair credit" idea is easiest to see on a **linear model**, where it collapses to a single line and there are **no coalitions to average**. Once it's obvious here, the companion worksheet **SHAP_ByHand** (a shallow *tree*) shows what changes the moment a model has **interactions**.

The model — one linear equation

A linear model scores a home (median value, in **units of 100,000 USD**) from three features — **I** = income (10k USD), **R** = rooms, **L** = latitude:

$$f(I, R, L) = \underbrace{1}_{\text{intercept } b} + \underbrace{2}_{\beta_I} I + \underbrace{3}_{\beta_R} R + \underbrace{1}_{\beta_L} L.$$

The **baseline** is the **average** of each feature (that's what "normal" means here); the **home** is the row we're explaining:

feature	baseline = mean $\bar{x}_i$	this home x_i
I (income, 10k USD)	3	8
R (rooms)	2	5
L (latitude)	4	6

The one formula — no coalitions needed

For a **linear** model, a feature's SHAP value is just:

$$\boxed{\phi_i = \beta_i(x_i - \bar{x}_i)} \quad (\text{"coefficient"} \times \text{how far this home is from average})$$

and the **base value** is the model at the averages, $f(\bar{x})$ — which is exactly the **average prediction**.

Why there are no coalitions here: a linear model has **no interactions** — a feature's effect (β_i) doesn't depend on the other features. So no matter what order the features "arrive" in, feature i always adds the same amount, $\beta_i(x_i - \bar{x}_i)$. Averaging identical numbers just gives that number back — so the whole Shapley machinery collapses to one term. (*On the tree worksheet, interactions break this, and that's when the coalition average earns its keep.*)

Tasks

1. **Base value.** Compute $f(\bar{x})$ — the model at the baseline means.
 2. **Prediction.** Compute $f(x)$ — the model at this home's values.
 3. **SHAP values.** Compute ϕ_I, ϕ_R, ϕ_L with $\phi_i = \beta_i(x_i - \bar{x}_i)$.
 4. **Efficiency (local accuracy).** Check that **base value** + $\phi_I + \phi_R + \phi_L =$ **prediction**.
 5. **Notice.** Each ϕ_i is exactly *one term of the equation*, measured from the average. Write one sentence saying why that makes SHAP "obvious" for a linear model.
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